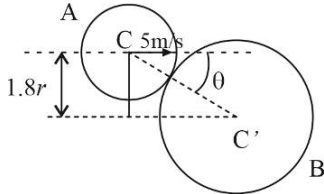


Solutions to JEE Advanced Booster Test – 2 | 2022 | Code A

[PHYSICS]

1.(BD) $\sin \theta = \frac{1.8r}{3r} = \frac{3}{5}$

Velocity of centre of A along common normal $cc' = 5 \cos \theta = 4 \text{ m/s}$



Just before collision

Component of velocity perpendicular to $cc' = 5 \sin \theta = 3 \text{ m/s}$

After collision, velocity of A along common normal changes but the component perpendicular to it remains same

Also B will gain velocity only along the common normal.

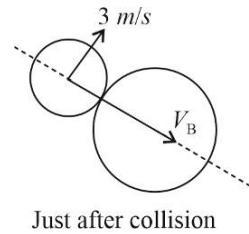
It is observed that A and B move perpendicular to each other, so velocity of A along common normal should become zero.

$$\therefore V_A = 3 \text{ m/s}$$

To find V_B , apply momentum conservation

$$m_A(4) = m_B V_B \Rightarrow V_B = 1 \text{ m/s}$$

$$e = \frac{1-0}{4-0} = 0.25$$



Just after collision

2.(BC) Let the velocity of the CM of the cylinder be v_{CM}

Since the velocity of the point of the cylinder in contact with the surface must be zero, $v_{CM} = R\omega$

And, since the velocity of the point of the cylinder in contact with the plank must be equal to the velocity of the plank, $v_{CM} + R\omega = v$

Solving we get, $v_{CM} = \frac{v}{2}$ and $\omega = \frac{v}{2R}$

Now, the kinetic energy of the cylinder,

$$K = \frac{1}{2} M v_{CM}^2 + \frac{1}{2} \left(\frac{1}{2} M R^2 \right) \omega^2 = \frac{1}{2} M \left(\frac{v}{2} \right)^2 + \frac{1}{4} M R^2 \left(\frac{v}{2R} \right)^2 = \frac{3}{16} M v^2$$

3.(BD) Let the mass of the particle be m , the radius of the particle's path be r and let its speed be v

At $t = \frac{T}{4}$, the particle has completed one quarter of the circle and is situated at the point on the +Y axis,

with its velocity tangential to the circular path, i.e. towards the -X direction.

Therefore, $L_O = mvr$ and $L_S = mv(2r) = 2mvr$

But $K = \frac{1}{2} mv^2$ and $T = \frac{2\pi r}{v}$

$$\Rightarrow KT = \pi mvr$$

Therefore, $L_O = \frac{KT}{\pi}$ and $L_S = \frac{2KT}{\pi}$

Also, we can see quite clearly that both $\vec{L}_O$ and $\vec{L}_S$ are towards the +Z direction.

4.(ABD) Before the string breaks, the acceleration of both blocks,

$$a = \left(\frac{2.1 - 1.9}{2.1 + 1.9} \right) g = 0.5 \text{ m/s}^2 \text{ upward for block A and downward for block B}$$

Therefore, at $t = 1.0 \text{ s}$, the velocities of the blocks are: (taking upward positive)

$$v_{A1} = (0.5)(1) = 0.5 \text{ m/s}$$

$$v_{B1} = (-0.5)(1) = -0.5 \text{ m/s}$$

After the string breaks, acceleration of both blocks is g downwards

So, at $t = 1.1 \text{ s}$, the velocities of the blocks are:

$$v_{A2} = 0.5 + (-10)(0.1) = -0.5 \text{ m/s}$$

$$v_{B2} = -0.5 + (-10)(0.1) = -1.5 \text{ m/s}$$

Now, between $t = 0$ and $t = 1.1 \text{ s}$, the total displacements of the two blocks are:

$$s_A = \left(\frac{1}{2} (0.5)(1)^2 \right) + \left((0.5)(0.1) + \frac{1}{2} (-10)(0.1)^2 \right) = 0.25 \text{ m} = 25 \text{ cm}$$

$$s_B = \left(\frac{1}{2} (-0.5)(1)^2 \right) + \left((-0.5)(0.1) + \frac{1}{2} (-10)(0.1)^2 \right) = -0.35 \text{ m} = -35 \text{ cm}$$

5.(AC) We know that the common tangent of the two bodies during the collision is along surface of the inclined plane.

The component of the velocity of the particle along the common tangent must remain unchanged during the collision.

$$\text{Therefore, } v \sin \theta = u \cos \theta \Rightarrow v = u \cot \theta$$

Also, since there is no external impulse on the system of the particle + inclined plane in the horizontal direction, we can conserve momentum along this direction.

$$\text{Therefore, } mu = (4m)v_p \Rightarrow v_p = \frac{u}{4}$$

6.(AD) Since the velocity of the bottom-most point at $t = 0$ is towards right, kinetic friction $f_K = \mu mg$ acts towards left

Therefore, the acceleration of the centre of mass,

$$a_{CM} = \frac{f_K}{m} = \mu g \quad (\text{towards left})$$

And, the angular acceleration of the cylinder,

$$\alpha = \frac{f_K R}{\frac{1}{2} m R^2} = \frac{2 f_K}{m R} = \frac{2 \mu g}{R} \quad (\text{clockwise})$$

So, taking rightward positive and clockwise positive, at any general time instant t ,

$$v_{CM} = v_0 - \mu g t \quad \text{and} \quad \omega = -\omega_0 + \left(\frac{2 \mu g}{R} \right) t$$

Pure rolling begins when $v_{CM} = R\omega$

(Note carefully that we can write this because we have taken rightward positive for v_{CM} and clockwise positive for ω)

$$\text{So, } v_0 - \mu g t = R \left(-\omega_0 + \left(\frac{2 \mu g}{R} \right) t \right) \Rightarrow t = \frac{v_0 + R\omega_0}{3 \mu g} = \frac{1 + (0.1)(50)}{3(0.1)(10)} = 2 \text{ s}$$

$$\text{Therefore, at } t = 2 \text{ s, } v_{CM} = v_0 - \mu g t = 1 - (0.1)(10)(2) = -1 \text{ m/s}$$

$$\text{So, } v_{CM} = 1 \text{ m/s towards left}$$

And, $\omega = -\omega_0 + \left(\frac{2\mu g}{R}\right)t = -50 + \left(\frac{2(0.1)(10)}{0.1}\right)(2) = -10 \text{ rad/s}$

So, $\omega = 10 \text{ rad/s}$ anti-clockwise

Once pure rolling begins, v_{CM} and ω remain constant at the values they had at $t = 2 \text{ s}$.

7.(C) Gain in kinetic energy = Loss in potential energy $K = mg\left(\frac{L}{4}\right) + (2m)g\left(\frac{3L}{4}\right) = \frac{7}{4}mgL$

8.(C) Moment of inertia of the rod about the pivot,

$$I = \frac{1}{3}m\left(\frac{L}{2}\right)^2 + \left(\frac{1}{12}(2m)\left(\frac{L}{2}\right)^2 + (2m)\left(\frac{3L}{4}\right)^2\right) = \frac{5}{4}mL^2$$

So, if the angular velocity of the rod in the vertical position is ω ,

$$\frac{1}{2}I\omega^2 = K \Rightarrow \frac{1}{2}\left(\frac{5}{4}mL^2\right)\omega^2 = \frac{7}{4}mgL \Rightarrow \omega = \sqrt{\frac{14g}{5L}}$$

9.(D) Let the velocity of the spaceship after the projectile is fired be v_f

Conserving momentum,

Initial momentum = Final momentum $Mv = \left(\frac{199}{200}\right)Mv_f + \left(\frac{M}{200}\right)(4v) \Rightarrow v_f = \left(\frac{196}{199}\right)v$

10.(A) Let the mass of the object X be M_x

Conserving momentum,

Initial momentum = Final momentum

$$\left(\frac{M}{200}\right)(4v) = \left(\frac{M}{200} + M_x\right)\left(\frac{v}{3}\right) \Rightarrow M_x = \left(\frac{11}{200}\right)M$$

11.(D) Work done by a force on a particle is given by, $W = \int \vec{F} \cdot d\vec{r}$, where $d\vec{r} = dx \hat{i} + dy \hat{j}$

(p) $W = \int \vec{F} \cdot (dx \hat{i} + dy \hat{j}) = \int (\hat{i} + \hat{j}) \cdot (dx \hat{i} + dy \hat{j}) = \int_{-1}^1 dx + \int_{-1}^1 dy = 4 \text{ J}$

(q) $W = \int \vec{F} \cdot (dx \hat{i} + dy \hat{j}) = \int (\hat{i} - 2\hat{j}) \cdot (dx \hat{i} + dy \hat{j}) = \int_{-1}^1 dx + (-2) \int_{-1}^1 dy = 2 - 4 = -2 \text{ J}$

(r) $W = \int \vec{F} \cdot (dx \hat{i} + dy \hat{j}) = \int (3y^2 \hat{i}) \cdot (dx \hat{i} + dy \hat{j}) = \int_{-1}^1 3y^2 dx$

Here, it seems that we are stuck because the integral cannot be solved as it has two variables x and y . But we can use the fact that the particle is moving on the straight line $y = x$.

So, $W = \int_{-1}^1 3x^2 dx = 2 \text{ J}$

(s) $W = \int \vec{F} \cdot (dx \hat{i} + dy \hat{j}) = \int (x\hat{i} + y\hat{j}) \cdot (dx \hat{i} + dy \hat{j}) = \int_{-1}^1 x dx + \int_{-1}^1 y dy = 0$

12.(B) Since $m_1 \sin \theta > m_2$, in the absence of friction block A slips down the plane. Therefore, for equilibrium, friction must act up the plane. Let its magnitude be f .

Then, in equilibrium, $f = m_1 g \sin \theta - m_2 g$

Now, for equilibrium to be possible, $f \leq \mu m_1 g \cos \theta$

Therefore, $m_1 g \sin \theta - m_2 g \leq \mu m_1 g \cos \theta$

So, $\mu \geq \frac{m_1 \sin \theta - m_2}{m_1 \cos \theta}$

We know that $\frac{m_1 \sin \theta - m_2}{m_1 \cos \theta} = \frac{(2)(0.6) - 1}{2(0.8)} = \frac{1}{8}$

Therefore, $\mu \geq \frac{1}{8}$ for equilibrium

In the case when $\mu < \frac{1}{8}$, block A accelerates down with acceleration,

$$a = \frac{m_1 g \sin \theta - m_2 g - \mu m_1 g \cos \theta}{m_1 + m_2} = \frac{(2)(10)(0.6) - (1)(10) - \mu(2)(10)(0.8)}{2 + 1} = \frac{2}{3} - \frac{16}{3}\mu$$

So, when $\mu = 0.05$, $a = \frac{2}{3} - \frac{16}{3}(0.05) = 0.4 \text{ m/s}^2$

And, when $\mu = 0.1$, $a = \frac{2}{3} - \frac{16}{3}(0.1) = 0.13 \text{ m/s}^2$

13.(7.50) Balancing torque about the point A,

Clockwise torque = Anti-clockwise torque

$$\Rightarrow R(L \sin \theta) = mg \left(\frac{L}{2} \cos \theta \right) \Rightarrow R = \frac{mg}{2} \cot \theta$$

Therefore, $R = \frac{(2)(10)}{2} \left(\frac{3}{4} \right) = 7.5 \text{ N}$

14.(1) Let the velocity of the block after it has moved down by a distance x be v

Conserving energy,

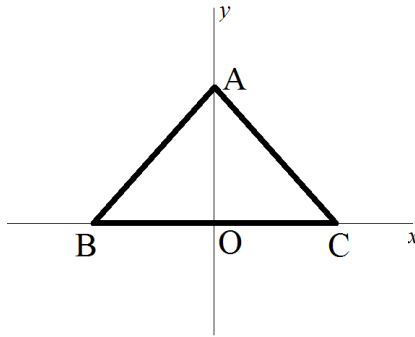
Loss in gravitational PE = Gain in kinetic energy + Gain in elastic PE

$$\Rightarrow mgx = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \Rightarrow v^2 = \frac{2mgx - kx^2}{m}$$

We are given that $m = 1 \text{ kg}$ $k = 100 \text{ N/m}$ and $x = 0.1 \text{ m}$

Therefore, $v^2 = \frac{2(1)(10)(0.1) - (100)(0.1)^2}{1} = 1 \Rightarrow v = 1 \text{ m/s}$

15.(2.75)



Taking the origin at the mid-point of BC and the axes as shown, we can see that A, B and C have the coordinates:

$$A(0, 4L) \quad B(-3L, 0) \quad C(3L, 0)$$

Since the CM of each side of the triangle is at the mid-point of that side, the CM of the sides have the coordinates:

$$\text{CM of AB} \equiv \left(-\frac{3L}{2}, 2L \right) \quad \text{CM of BC} \equiv (0, 0) \quad \text{CM of AC} \equiv \left(\frac{3L}{2}, 2L \right)$$

Also, the mass of each side is proportional to its length, so let the masses of AB, BC and AC be $5M$, $6M$ and $5M$ respectively.

Therefore, for the triangle,

$$X_{CM} = \frac{5M\left(-\frac{3L}{2}\right) + 6M(0) + 5M\left(\frac{3L}{2}\right)}{16M} = 0$$

$$Y_{CM} = \frac{5M(2L) + 6M(0) + 5M(2L)}{16M} = \frac{5}{4}L$$

[Of course, we did not need to calculate X_{CM} because we know that the triangle is symmetric about the line AO and hence the CM must lie on it]

$$\text{Distance of the CM from A} = 4L - \frac{5}{4}L = \frac{11}{4}L = (2.75)L$$

16.(3) Let the acceleration of the blocks B and C be a

We know that B will accelerate upwards and C will accelerate downwards

$$\text{Therefore,} \quad T - 1g = 1a \quad \text{and} \quad 3g - T = 3a$$

$$\text{Solving, we get} \quad a = 5 \text{ m/s}^2 \quad \text{and} \quad T = 15 \text{ N}$$

The block A is in equilibrium

$$\text{Therefore,} \quad 2T = m_A g \quad \Rightarrow \quad m_A = 3 \text{ kg}$$

17.(0.60) Conserving angular momentum about the hinge point,

Initial angular momentum = Final angular momentum

$$\Rightarrow \quad mvL = \left(\frac{1}{3}(2m)L^2 + mL^2 \right) \omega \quad \Rightarrow \quad \omega = \frac{3v}{5L}$$

18.(17) The given body can be considered the superposition of:

(a) a disc of radius $2R$ and mass per unit area σ_0 , and

(b) a disc of radius R and mass per unit area σ_0

Now, moment of inertia of a disc of radius R about its diameter,

$$I = \frac{1}{4}mR^2 = \frac{1}{4}(\sigma_0 \pi R^2)R^2 = \frac{1}{4}\pi\sigma_0 R^4$$

Therefore, total moment of inertia of the given body about a diameter,

$$I_{Total} = \frac{1}{4}\pi\sigma_0 R^4 + \frac{1}{4}\pi\sigma_0 (2R)^4 = \frac{17}{4}\pi\sigma_0 R^4$$

So, the kinetic energy of the body about the diameter is.

$$K = \frac{1}{2}I_{Total}\omega^2 = \frac{17}{8}\pi\sigma_0 R^4 \omega^2$$

[CHEMISTRY]

1.(ABC) $\text{:}\ddot{\text{I}}-\ddot{\text{I}}-\ddot{\text{I}}\text{:}$ Linear sp^3d hybridization

$\begin{array}{c} \ddot{\text{O}} \\ // \quad \backslash \\ \text{:}\ddot{\text{O}}\text{:} \quad \text{:}\ddot{\text{O}}\text{:} \end{array}$ sp^2 hybridization Bent shape

$^{2-}\text{C}=\text{C}=\text{C}^{2-}$ sp hybridization Linear

order of p character $\text{C}_3^{4-} < \text{I}_3^- < \text{O}_3$

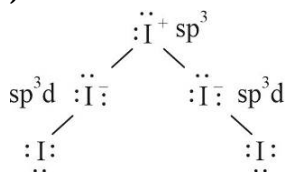
2.(AC) Graphite is thermodynamically most stable state of carbon and thus $\Delta_f H^\circ$ (graphite) = 0

3.(AC) (B) B_2 has only 1π bond acc to MOT but HF has 1 σ bond

(D) Boiling point $\text{HF} < \text{H}_2\text{O}$

4.(BCD) Gas, liquid and solid coexist at triple point. Ideal gas can never be liquified thus no critical point for ideal gas.

5.(ACD)



It has 4 σ bond

Total lone pairs = 14

6.(ABCD) Internal energy (U) and Enthalpy (H) are state function. Whereas work (w) and heat (q) are path functions. $q + w = \Delta U$ (first law of thermodynamics)

7.(B) Gas having strongest intermolecular forces will have smallest molar volume. In NH_3 there is H-bonding.

8.(B) Stronger the intermolecular force acting among the molecule, lower will be its vapour pressure.

9.(A) Enthalpy change when 1 mol of a compound is being formed from its constituents in this respective reference state is called enthalpy of formation.

10.(B) $2\text{C (graphite)} + 3\text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_6(\text{g}) \quad \Delta H_f^\circ = -84.7$

(i) $\frac{1}{2}\text{H}_2(\text{g}) \longrightarrow \text{H}(\text{g}) \quad \Delta H_f^\circ = 217.9 \text{ kJ/mol}$

(ii) $\text{C (graphite)} \longrightarrow \text{C}(\text{g}) \quad \Delta H_f^\circ = 718.4 \text{ kJ/mol}$

(iii) $\text{C-H}(\text{g}) \longrightarrow \text{C}(\text{g}) + \text{H}(\text{g}) \quad \Delta H^\circ = 416 \text{ kJmol}^{-1}$

(iv) $\text{C-C}(\text{g}) \longrightarrow 2\text{C}(\text{g}) \quad \Delta H^\circ = ?$

$\Delta H_f^\circ(\text{C}_2\text{H}_6) = 2(\text{ii}) + 6(\text{i}) - 6(\text{iii}) - (\text{iv})$

$\Rightarrow -84.7 = 2(718.4) + 6(217.9) - 6(416) - (\text{iv})$

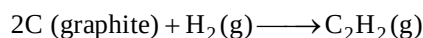
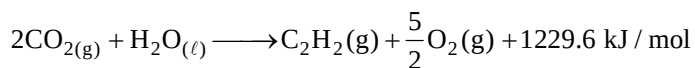
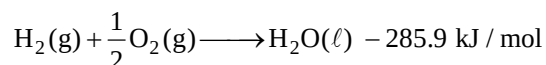
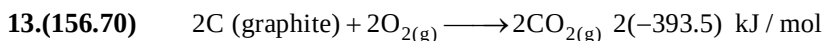
$\Rightarrow (\text{iv}) = \Delta H_{\text{bond (C-C)}}^\circ = 333 \text{ kJ/mol}$

11.(A)

	Property		Type/ Effect of temperature or pressure
I.	Surface Tension of water	P	Intensive property, Decrease with increase in temperature
II.	Boiling Point of water	Q	Intensive property, Decrease with increase in pressure
III.	Density of water	R	Intensive property, Decrease with increase in temperature
IV.	Aqueous Tension of water	S	Intensive property, Increase with increase in temperature

12 (B)

	Molecular species	Properties
I	H ₂ O	sp ³ hybridized central atom, polar, not paramagnetic, $\Delta H_f^0 \neq 0$
II	H ₂ O ₂	sp ³ hybridized central atom, polar, not paramagnetic, $\Delta H_f^0 \neq 0$
III	HO ₂	sp ³ hybridized central atom, polar, paramagnetic, $\Delta H_f^0 \neq 0$
IV	O ₂	Paramagnetic, $\Delta H_f^0 = 0$
V	O ₃	sp ² hybridized central atom, polar, not paramagnetic, $\Delta H_f^0 \neq 0$



$$\Delta_f H^\circ(C_2H_2) = 2(-393.50) + (-285.9) + 1229.6$$

$$= 156.70 \text{ kJ/mol}$$

14.(25.38) $\Delta H = mc\Delta T$

$$= (10 + 100)4.184(24 - 17.11) = 3171.05 \text{ J} = 3.17 \text{ kJ}$$

$$10 \text{ g of } NH_4NO_3 = 3.17 \text{ kJ}$$

$$80.05 \text{ g of } NH_4NO_3 = \frac{3.17}{10} \times 80.05 = 25.38 \text{ kJ mol}^{-1}$$

15.(6) $n_i = 30 \text{ mol} \quad n_f = ?$

$$T_i = 47^\circ\text{C} \quad T_f = 27^\circ\text{C}$$

$$P_i = P \text{ atm} \quad P_f = (0.75)P \text{ atm}$$

Since volume of canister is constant

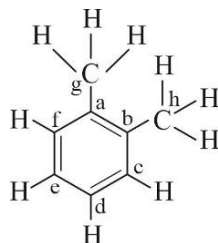
$$\therefore \frac{nT}{P} = \text{Constant}$$

$$\Rightarrow \frac{n_i T_i}{P_i} = \frac{n_f T_f}{P_f} \Rightarrow \frac{30 \text{ mol} \times 320 \text{ K}}{P} = \frac{n_f \times 300 \text{ K}}{0.75P} \Rightarrow n_f = 32 \times 0.75 = 24 \text{ mol}$$

$$\text{Amount of } N_2 \text{ leaked} = 6 \text{ mol}$$

16.(7) B₂, O₂, Na, KO₂, NO₂, NO, ClO₂ are paramagnetic substances and appear heavier if weight is measured in a magnetic field.

17.(38) ($x_1 = 14$, $x_2 = 6$, $x_3 = 18$)

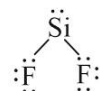
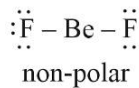
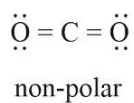
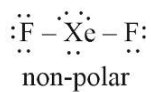
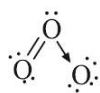


a, b, c, d, e, f = sp^2 hyb C atoms

g, h = sp^3 hyb C atoms

$$x_1 + x_2 + x_3 = 38$$

18.(1) $:\ddot{\text{N}}^{\ominus} = \text{N}^{\oplus} = \ddot{\text{O}}:$ linear and polar



1.(AB) Find the equation of the lines through (2, 3) and parallel to the bisectors of the given lines, the slopes of the bisectors being $-\frac{1}{3}$ and 3.

2.(BC) T_{r+1} in $\left(x^4 + \frac{1}{x^3}\right)^{15}$

$$\text{is } T_{r+1} = {}^{15}C_r (x^4)^{15-r} x^{-3r} = {}^{15}C_r x^{60-7r}$$

$$60 - 7r = 0$$

r is not an integer

Hence, no term independent of x .

(a) is not correct.

Again coefficient of 8th term i.e., $r = 7$ is ${}^{15}C_7$ and 9th term i.e., $r = 8$ is ${}^{15}C_8$

$$\therefore {}^{15}C_7 = {}^{15}C_8$$

$$\text{Again, } 60 - 7r = 32$$

$$r = 4$$

$$60 - 7r = -17$$

$$r = 11$$

$$\therefore \text{Coefficient of } x^{32} = {}^{15}C_4 \text{ and } x^{-17} = {}^{15}C_{11}$$

$$\text{If } x = \sqrt{2}, T_{r+1} = {}^{15}C_r \cdot 2^{\frac{60-7r}{2}}$$

Hence rational terms occurs if

$$r = 0, 2, 4, 6, 8, 10, 12, 14 \quad \therefore \quad \text{(B), (C) are correct.}$$

3.(ABC) x, y, z friend

a times, b times, c times

$$a, b, c \equiv (1, 1, 4), (1, 2, 3), (2, 2, 2)$$

$$\text{Numbers of ways} = \frac{6!3!}{4!1!1!2!} + \frac{6!3!}{1!2!3!} + \frac{6!3!}{1!2!3!} = 90 + 360 + 90 = 540$$

$$N = 540 = 2^2 \cdot 3^3 \cdot 5^1$$

(A) Divisors divisible by 5 is : $3 \times 4 \times 1 = 12$

$$\text{(B) Sum} = (2^0 + 2^1 + 2^2)(3^0 + 3^1 + 3^2 + 3^3)(5^0 + 5^1) = 7 \times 40 \times 6 = 1680$$

$$\text{(C) } 2^{n-1} = 2^2 = 4$$

(D) Even and not divisible by 4

$$\therefore 1 \times 4 \times 2 = 8$$

4.(BC) $a = 36; b = 18; c = 36$

$$a = \frac{4!}{2!} \times \frac{3!}{2!} = 12 \times 3 = 36;$$

$$b = \frac{3!}{2!} \times \frac{3!}{2!} \times 2! = 18;$$

$$c = \frac{4!}{2!} + \frac{4!}{2!} \times 2! = 12 + 24 = 36$$

5.(AB) Let inclination of AC is θ

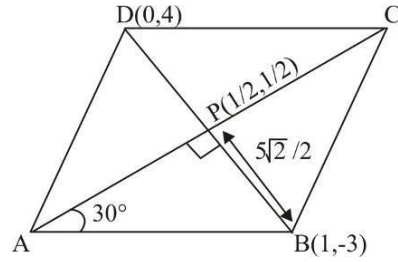
$$\tan \theta = \frac{1}{7}, \sin \theta = \frac{1}{5\sqrt{2}}, \cos \theta = \frac{7}{5\sqrt{2}}$$

$$AP = \frac{5}{2}\sqrt{2} \cot 30^\circ = \frac{5\sqrt{2}}{2}\sqrt{3}$$

$$A = \left(\frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2} \frac{\sqrt{7}}{5\sqrt{2}}, \frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2} \frac{1}{5\sqrt{2}} \right)$$

$$\text{Or } \left(\frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2} \frac{7}{5\sqrt{2}}, \frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2} \frac{1}{5\sqrt{2}} \right)$$

$$A = \left(\frac{1}{2} - \frac{7\sqrt{3}}{2}, \frac{1}{2} - \frac{\sqrt{3}}{2} \right) \text{ or } \left(\frac{1}{2} + \frac{7\sqrt{3}}{2}, \frac{1}{2} + \frac{\sqrt{3}}{2} \right)$$



6.(AC) $S_n = {}^{2n}C_{n-1} \quad \left| \quad S_{n+1} = {}^{2n+2}C_n \right.$
 $= \frac{(2n)!}{(n-1)!(n+1)!} \quad \left| \quad = \frac{(2n+2)!}{n!(n+2)!} \right.$

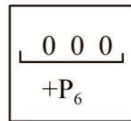
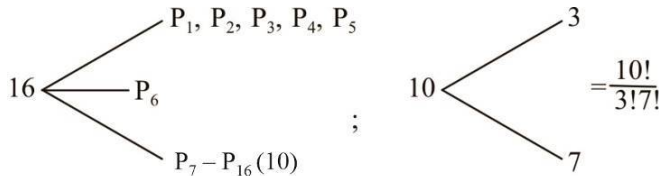
$$\frac{S_{n+1}}{S_n} = \frac{(2n+2)!}{n!(n+2)!} \times \frac{(n-1)!(n+1)!}{(2n)!} = \frac{(2n+2)(2n+1)}{n \times (n+2)}$$

as n approaches infinity $\frac{S_{n+1}}{S_n} = \frac{n^2 \left(2 + \frac{2}{n} \right) \left(2 + \frac{1}{n} \right)}{n^2 \left(1 + \frac{2}{n} \right)} = \frac{2 \times 2}{1} = 4$

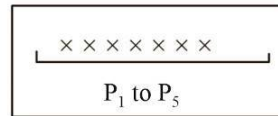
Similarly for $\frac{S_{n+3}}{S_n}$.

7.(A) Number of ways $= \frac{(16)!}{(4!)^4 4!} = \frac{2^8 \cdot 8! \prod_{r=1}^8 (2r-1)}{4! \cdot 4! \cdot 4! \cdot 4! \cdot 4!} = \frac{2^8 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot \lambda}{(24)(24)(24)(24)} = \frac{2^9 \cdot 35 \cdot \lambda}{(24) \cdot (24) \cdot (24)}$
 $= \frac{35}{27} \prod_{r=1}^8 (2r-1)$

8.(D)

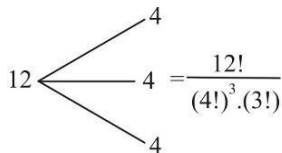


4 players



12 players

Now,



$$\text{Total ways} = \frac{(10)!}{3!7!} \cdot \frac{(12)!}{(4!)^3 \cdot 3!} = \left(\frac{12!}{(4!)^3} \right) \cdot 20 \Rightarrow k = 20$$

9.(D) We have $f(\alpha, \beta) \leq 0$

$$\left| \frac{\beta}{\alpha} - \frac{3}{2} \right| + (3\alpha - 2\beta)^6 - \sqrt{e\alpha + 2\beta - 2e - 6}$$

$$\frac{\beta}{\alpha} - \frac{3}{2} = 0$$

$$\beta = \frac{3\alpha}{2} \text{ and } e\alpha + 2\beta - 2e - 6 = 0$$

$$(\alpha - 2)e + 2(\beta - 3) = 0$$

Hence $\alpha = 2$ and $\beta = 3$

Let equation of straight line be $\frac{x}{\cos \theta} = \frac{y}{\sin \theta} = r$

Let $OA = r_1, OB = r_2$ and $OC = r_3$

Then $A(r_1 \cos \theta, r_1 \sin \theta), B(r_2 \cos \theta, r_2 \sin \theta)$

And $C(r_3 \cos \theta, r_3 \sin \theta)$

A lies on $x = 2$

$$r_1 \cos \theta = 2$$

$$r_1 = \frac{2}{\cos \theta} \dots (i)$$

B lies on $y = 3$

$$r_2 \sin \theta = 3$$

$$r_2 = \frac{3}{\sin \theta} \dots (ii)$$

And C lies on $x + y = 8$

$$r_3 \cos \theta + r_3 \sin \theta = 8$$

$$r_3 = \frac{8}{\cos \theta + \sin \theta} \dots (iii)$$

$$r_1 r_2 r_3 = 48\sqrt{2}$$

$$\left(\frac{2}{\cos \theta} \right) \left(\frac{3}{\sin \theta} \right) \left(\frac{8}{\cos \theta + \sin \theta} \right) = 48\sqrt{2}$$

$$\sqrt{2} \cos \theta \sin \theta (\cos \theta + \sin \theta) = 1$$

$$\sin 2\theta \sin \left(\theta + \frac{\pi}{4} \right) = 1$$

Both $\sin 2\theta$ and $\sin \left(\theta + \frac{\pi}{4} \right)$ should be equal to 1.

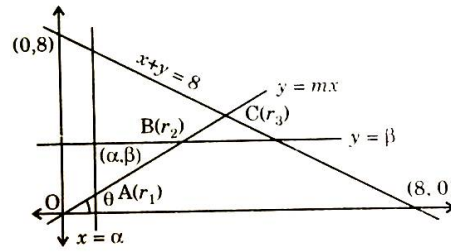
$$\text{So } \theta = \frac{\pi}{4}$$

$$\text{From (i), } r_1 = 2\sqrt{2}$$

$$\text{From (ii), } r_2 = 3\sqrt{2}$$

$$\text{From (iii), } r_3 = \frac{8}{\sqrt{2}} = 4\sqrt{2}$$

$$\text{Hence } OA + OB + OC = r_1 + r_2 + r_3 = 9\sqrt{2}$$



10.(A) The equation of line $\frac{x-0}{\cos \frac{\pi}{4}} = \frac{y-0}{\sin \frac{\pi}{4}} \Rightarrow x - y = 0$

11.(C) (I) $9^{103} = 9 \times (81)^{51} = 9 \times (80+1)^{51}$
 $= 9 \{ {}^{51}C_0 (80)^{51} + {}^{51}C_1 (80)^{50} + {}^{51}C_2 (80)^{49} + \dots + {}^{51}C_{50} (80) + 1 \}$
 $= 9(25k + (51 \times 80 + 1)) = 25\lambda + 36729 = 25\lambda + 25 \times 1469 + 4 = \lambda(25 + 1469) + 4$

Hence the remainder is 4

(II)
$$\sum_{r=1}^n \left(\sum_{r_1=0}^{r-1} {}^nC_r \cdot {}^rC_{r_1} \cdot 2^{r_1} \right)$$

$$= \sum_{r=1}^n {}^nC_r \left(\sum_{r_1=0}^{r-1} {}^rC_{r_1} \cdot 2^{r_1} \right) = \sum_{r=1}^n {}^nC_r \{ (1+2)^r - 2^r \} = \sum_{r=1}^n {}^nC_r 3^r - \sum_{r=1}^n {}^nC_r 2^r$$

$$= \{ (1+3)^n - 1 \} - \{ (1+2)^n - 1 \} = 4^n - 3^n \equiv x^n - y^n$$

 $\therefore x + y = 7$

(III) Sum of coefficients in the expansion of $(1+2x)^n$ is 6561, when $x = 1$

$$3^n = 6561 \Rightarrow 3^n = 3^8 \Rightarrow n = 8$$

Now, $\frac{T_{r+1}}{T_r} = \frac{{}^8C_r (2x)^r}{{}^8C_{r-1} (2x)^{r-1}} = \frac{9-r}{r} \times 2x$

$$\frac{T_{r+1}}{T_r} = \frac{9-r}{r}$$

$$\therefore \frac{T_{r+1}}{T_r} > 1 \Rightarrow \frac{9-r}{r} > 1$$

$$9-r > r ; \quad r < \frac{9}{2} \quad \text{Hence } 5^{\text{th}} \text{ term is greatest term}$$

(IV) Let $(6\sqrt{6}+14)^{2n+1} = P = I + F$

Where, I is as positive integer and $0 < F < 1$

$$(6\sqrt{6}-14)^{2n+1} = F', \text{ where } 0 < F' < 1$$

Subtracting we get, $2 \left[{}^{2n+1}C_1 (6\sqrt{6})^{2n} \times 14 + {}^{2n+1}C_3 (6\sqrt{6})^{2n-2} \times 14^3 + \dots \right]$
 $= I + F - F' \quad \therefore I + F - F' = \text{an even integer}$

Since $0 < F < 1$ & $0 < F' < 1$

Only possibility is $F - F' = 0$ or $F = F'$

$$\text{Also } PF = PF' = (6\sqrt{6}+14)^{2n+1} (6\sqrt{6}-14)^{2n+1} = (216-196)^{2n+1} = (20)^{2n+1} \equiv (k)^{2n+1}$$

$$\frac{k}{10} = \frac{20}{10} = 2$$

12.(C) $D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ and $D = 0$ is condition of concurrency

$$D = -(a^3 + b^3 + c^3 - 3abc) = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

(A) If $a + b + c = 0$

But $\sum a^2 \neq \sum ab$ i.e., a, b, c are not all equal, then $D = 0$

Hence lines are concurrent $\Rightarrow$ (q)

- (B) If $a + b + c = 0$ and $\sum a^2 = \sum ab \Rightarrow a = b = c \therefore a = 0; b = 0; c = 0$
 $\Rightarrow$ Lines becomes identical and of the form $0x + 0y + 0 = 0$
 Any ordered pair (x, y) will satisfy
 $\Rightarrow$ Complete xy plane $\Rightarrow$ (p)
- (C) If $a + b + c \neq 0$ and $\sum a^2 \neq \sum ab \Rightarrow a = b = c$
 Hence lines becomes neither identical nor coincident $\Rightarrow$ (r)
- (D) $\sum a^2 = \sum ab \Rightarrow a = b = c$
 Hence the system represents coincident lines $x + y + 1 = 0$

13.(2) Binomial coefficients of $2^{\text{nd}}, 3^{\text{rd}}, 4^{\text{th}}$ terms are ${}^m C_1, {}^m C_2, {}^m C_3$

They are in AP

$$2 \cdot {}^m C_2 = {}^m C_1 + {}^m C_3$$

$$\frac{2 \cdot m \cdot (m-1)(m-2)!}{2!(m-2)!} = \frac{m \cdot (m-1)!}{(m-1)!} + \frac{m \cdot (m-1)(m-2)(m-3)!}{3!(m-3)!}$$

upon solving we get, $m = 7$ and $m = 2$ (not acceptable)

$$\left[\sqrt{2^{\log_{10}(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_{10} 3}} \right]^7$$

$$T_6 = 21$$

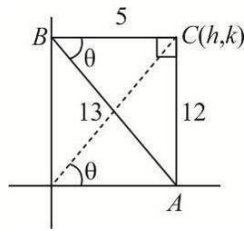
$$T_{5+1} = {}^7 C_5 \cdot 2^{\log_{10}(10-3^x)} \cdot 2^{(x-2)\log_{10} 3}$$

$$\text{Using } T_{r+1} = {}^n C_r x^{n-r} y^r$$

upon solving we get $x = 0$ and $x = 2$

answer is $x = 2$ (as larger value is asked)

14.(5)



$$\tan \theta = \frac{k}{h} = \frac{12}{5}$$

$$12x - 5y = 0 \Rightarrow k = 5$$

$$15.(8) \sum_{r=1}^{20} \left((r+1)^2 {}^{20} C_r - r^2 {}^{20} C_{r-1} \right) = 21^2 {}^{20} C_{20} - 1^2 {}^{20} C_0 = 440$$

16.(771) Case-1:

When no box remains empty it is equivalently distributing 10 coins in 5 beggar

$$\underbrace{0 \ 0 \ 0 \ 0 \ 0}_5 \underbrace{\bullet \ \bullet \ \bullet \ \bullet}_4 = {}^9 C_4 = 126$$

Case-2:

Exactly one is empty

$${}^5 C_1 \underbrace{0 \ 0 \ 0 \ 0 \ 0 \ 0}_6 \underbrace{\bullet \ \bullet \ \bullet}_3 = {}^9 C_3 = 420$$

Case-3:

Exactly two remains empty

$$({}^5C_2 \text{ -two adjacent}) {}^9C_2 \underbrace{0000000}_7 \bullet \bullet$$

$$(10-4) \times {}^9C_2$$

$$6 \times 36 = 216$$

Case-4:

Exactly three empty. There is only 1 way to select 3 if no two adjacent

$$\text{Hence } 1 \cdot {}^9C_1 = 900000000 \bullet$$

$$\text{Total} = 771 \text{ ways}$$

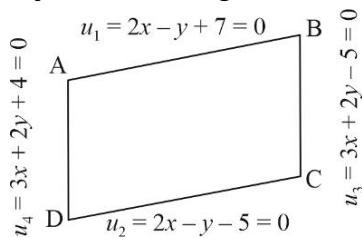
17.(296)

Category	Selection
$A=1$	$F"s=2$
$E=1$	$L;s=3$
$I=1$	$B=1$
$O=1$	$R=1$
	$M=1$

Total 12 letters

Category	Selection
(a) 3 alike + 2 different	$1(LLL) \cdot {}^8C_3 - {}^4C_3 = 56 - 4 = 52$
(b) 2 alike + 3 alike + 1 different	$1(LLLFF) \cdot {}^4C_1 = 4$
(c) 2 alike + 4 different	$2(LL \text{ or } FF) \cdot ({}^8C_4 - {}^4C_4) = 138$
(d) 2 alike + 2 alike + 2 different	$1(LL / FF) \cdot ({}^7C_2 - {}^3C_2) = 18$
(e) All six different	${}^9C_6 = {}^9C_3 = 84$
	Total ways = 296

18.(40) Equation of the diagonal AC



$$u_1 u_3 = u_2 u_4$$

$$(2x - y + 7)(3x + 2y - 5) = (3x + 2y + 4)(2x - y - 5)$$

$$-5(2x - y) + 7(3x + 2y) = -5(3x + 2y) + 4(2x - y) - 20$$

$$11x + 19y - 35 = -7x - 14y - 20$$

$$18x + 33y - 15 = 0 ; 6x + 11y - 5 = 0$$

$$\Rightarrow a = 6 \text{ and } b = 11$$

Equation of the other diagonal i.e. BD is $u_1 u_4 = u_2 u_3$

$$(2x - y + 7)(3x + 2y + 4) = (2x - y - 5)(3x + 2y - 5)$$

$$4(2x - y) + 7(3x + 2y) = -5(2x - y) - 5(3x + 2y) + 25$$

$$29x + 10y + 28 = -25x - 5y + 25$$

$$18x + 5y + 1 = 0$$

$$\Rightarrow p = 18 \text{ and } q = 5$$

$$\Rightarrow (a + b + p + q) = 6 + 11 + 18 + 5 = 40$$